Experience in the Same-Gender Environments and Low-Promotability Tasks

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Motivation

- Women often volunteer for or are assigned **low-promotability tasks (LPTs)**—tasks essential for group functioning but rarely rewarded.
- Examples: organizing meetings, taking notes, coordinating logistics.
- Such unequal allocation contributes to the **gender pay and promotion gap** (Babcock et al., 2017).

Research Question

- In same-gender groups, volunteering rates for LPTs are **similar** for men and women (Babcock et al., 2017).
- The gender gap **emerges primarily in mixed-gender settings**.
- Raises a key question about whether these behaviors are socially learned or internalized through prior experiences.

Q: Do experiences in same-gender environments shape women's willingness to volunteer for LPTs when they later move into mixed-gender groups? If so, how?

Conceptual Intuition for the Research Question

Babcock et al. (2017)'s main findings

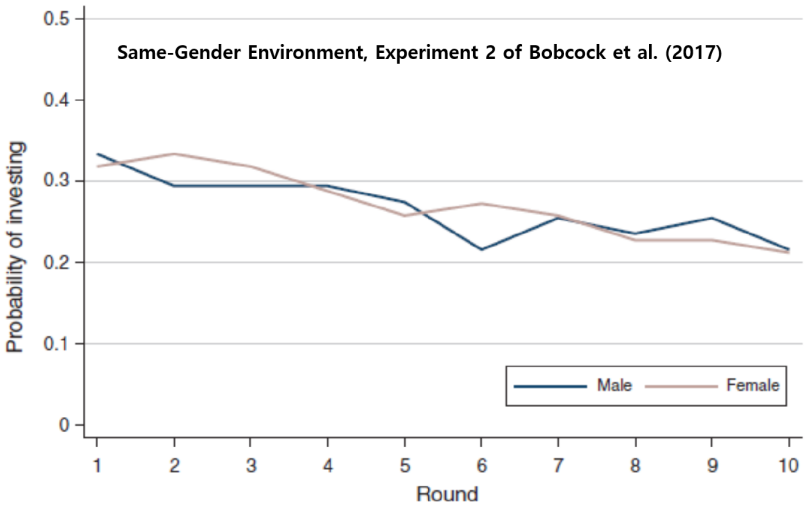


FIGURE 3. PROBABILITY OF INVESTING: EXPERIMENT 2

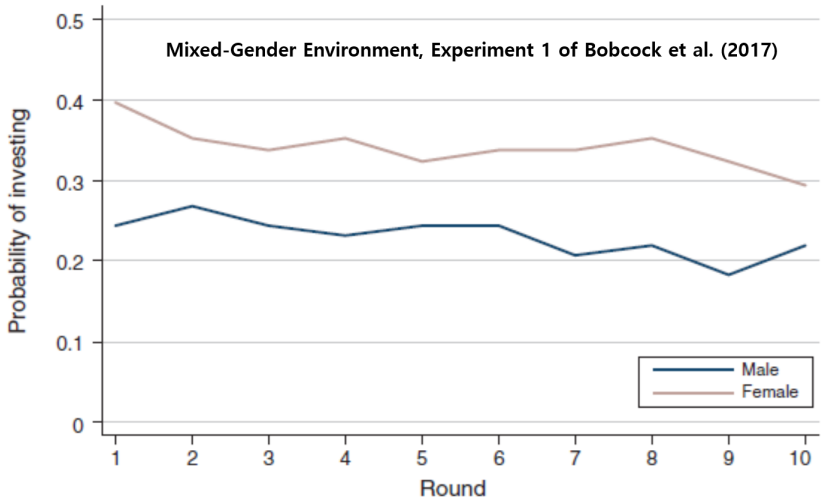
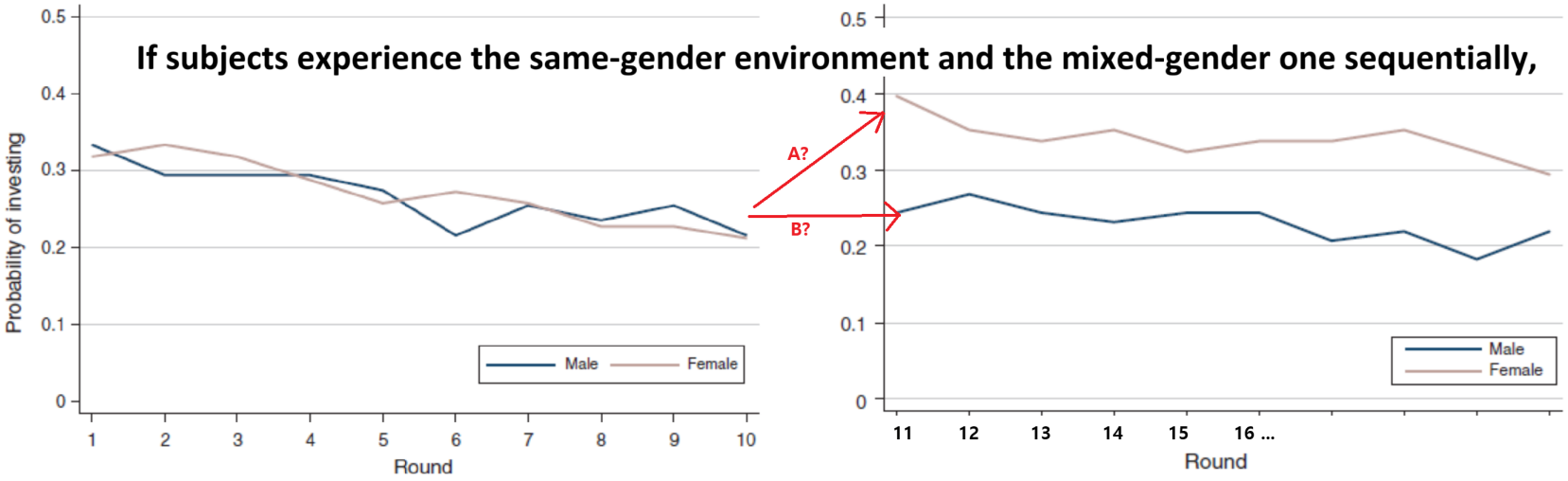


FIGURE 1. PROBABILITY OF INVESTING: EXPERIMENT 1

Conceptual Intuition for the Research Question

Our question



Would we observe (A) or (B)?

Related Literature

- **Gender difference in LPTs**: Babcock et al. (2017a, 2017b)
- **Replications and Field Evidence**: Villas-Boas et al. (2019), Banerjee and Mustafi (2025)
- **Task Assignment and Promotion Consequences**: Bircan et al. (2025)
- **LPTs in Higher Education**: Guarino and Borden (2017), O'Meara et al. (2017), Crimmins et al. (2023)
- **Single-Sex Schooling and Gendered Behavioral Patterns**: Lee et al. (2014), Booth and Nolen (2012), Booth et al. (2014), Laury et al. (2019), Ki et al. (2024)
- **Gender difference in avoiding strategic uncertainties**: Croson and Gneezy (2009), Cahudhuri (2011), Chang et al. (2024) Chang et al. (2024), Cason et al. (2022)

Gap: None of these studies examine **LPT behavior** or **carry-over into mixed-gender settings**.

Experimental Design

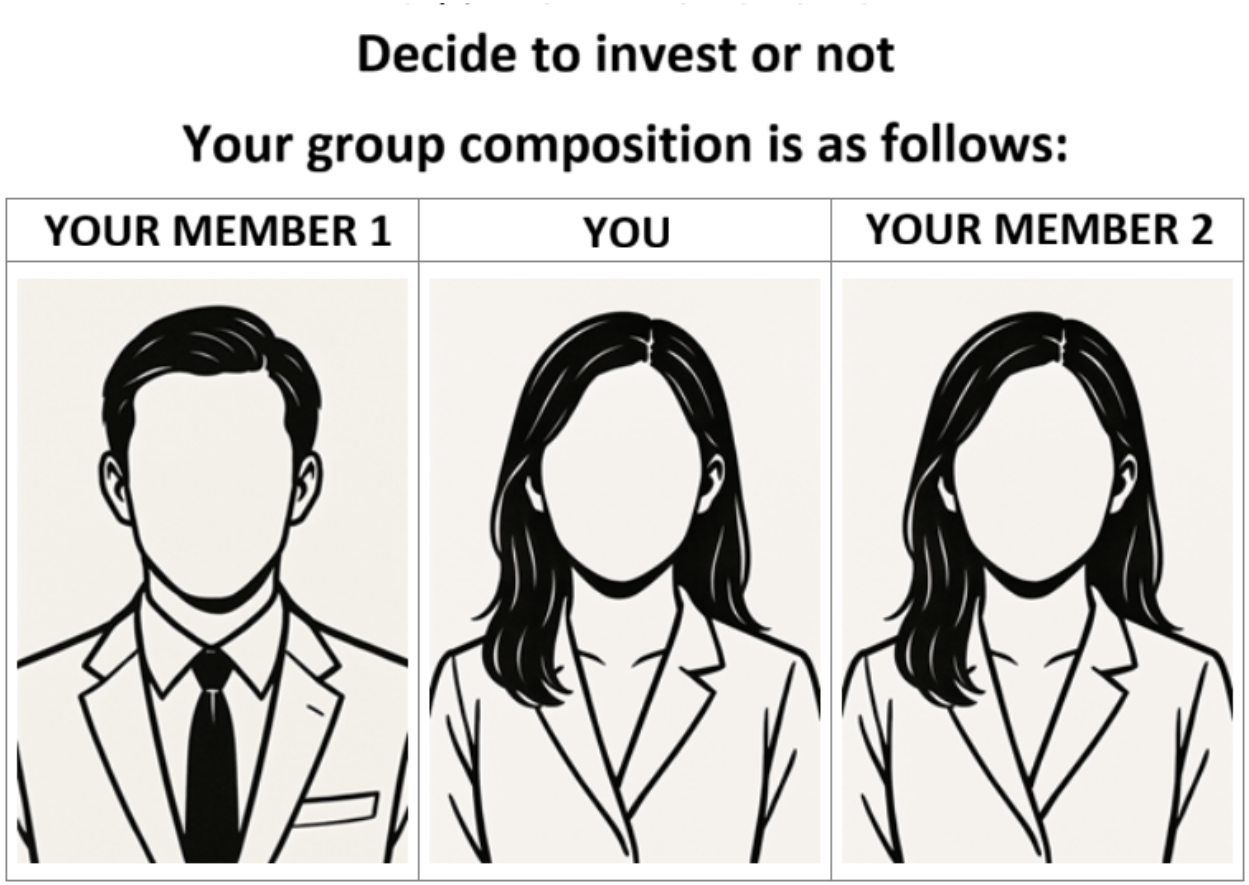
Within-subject two-stage laboratory experiment:

- Each participant plays 14 rounds of "investment game."
- In each round, a subject is newly assigned to a group of three. Each group member simultaneously decides to *invest or not*.
  - If no one invests, every member earns 100 tokens.
  - If one invests, that member earns 125 tokens, and the other two earn 200 tokens. (Investment = LPT)
  - If two or more decide to invest, randomly-selected one earns 125 tokens, and the other two earn 200 tokens.
- In the *SameFirst* treatment, subjects play 6 rounds of the game with same-gender members, and then play 8 rounds of the game with at least one member of different gender.
- To counterbalance a potential order effect, we also ran the *MixedFirst* treatment, where the 8 rounds of the mixed gender group game come first.

Challenge in Task Structure

Challenge: How to inform gender compositions without raising an experimenter demand effect?

Our approach: Silhouettes



Experiment Procedure

- Zoom-administered real-time online experiment
- LIONESS (Live Interactive ONline Experimental Server Software)
- University students aged 18 or over, in Korean
- 5 sessions (78 participants) for *SameFirst*, 4 sessions (72 participants) for *MixedFirst*.
- Twelve (6M6F) or eighteen (9M9F) subjects per session.
- Random regrouping with gender-composition constraints
- On average 15,700KRW (about 11 USD).
- Mobile payment credit (Naver Pay) corresponding to the cash value of one randomly-selected round

Testable Hypotheses

Hypothesis

Women volunteer for LPTs more than men in mixed-gender settings.

Hypothesis

Women volunteer for LPTs less in mixed-gender settings when they initially experience in same-gender settings.

Hypothesis

Women volunteer for LPTs less when they are with single-sex schooling backgrounds.

Hypothesis

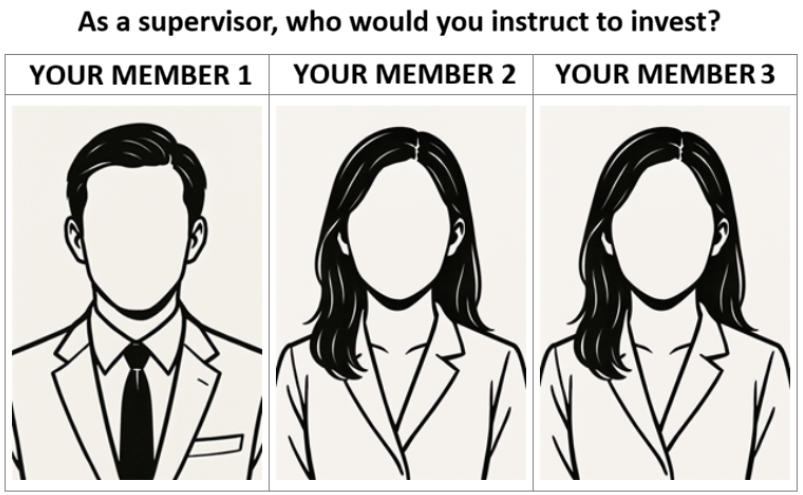
Subjects as supervisors would more assign an LPT to a female member in a mixed-gender group than a male member.

Experiment Procedure

- Collecting **gender**, birth year, and contact info when subjects sign up for participation.
- Turndown unmatched participants (e.g., when 11 males and 8 females showed up in the Zoom waiting room, 5 males and 2 females were turned down.)
- 14 rounds of investment game, no feedback
- Post-experiment survey collects:
  - Likert-scale questions, including:
    - "I feel pressure to invest,"
    - "I expect someone else with the same gender of mine would invest,"
    - "I volunteer a reluctant task if someone had to do,"
  - Types of the middle and high schools attended
  - Two hypothetical questions "assigning an investment" as a boss: One with 2M and 1F, and the other one with 1M and 2F. (See screenshots in the next slide.)

Two hypothetical questions

"If you were in a position as a supervisor of the group and had to instruct one of the group members to make an investment, which of the following would you choose?"

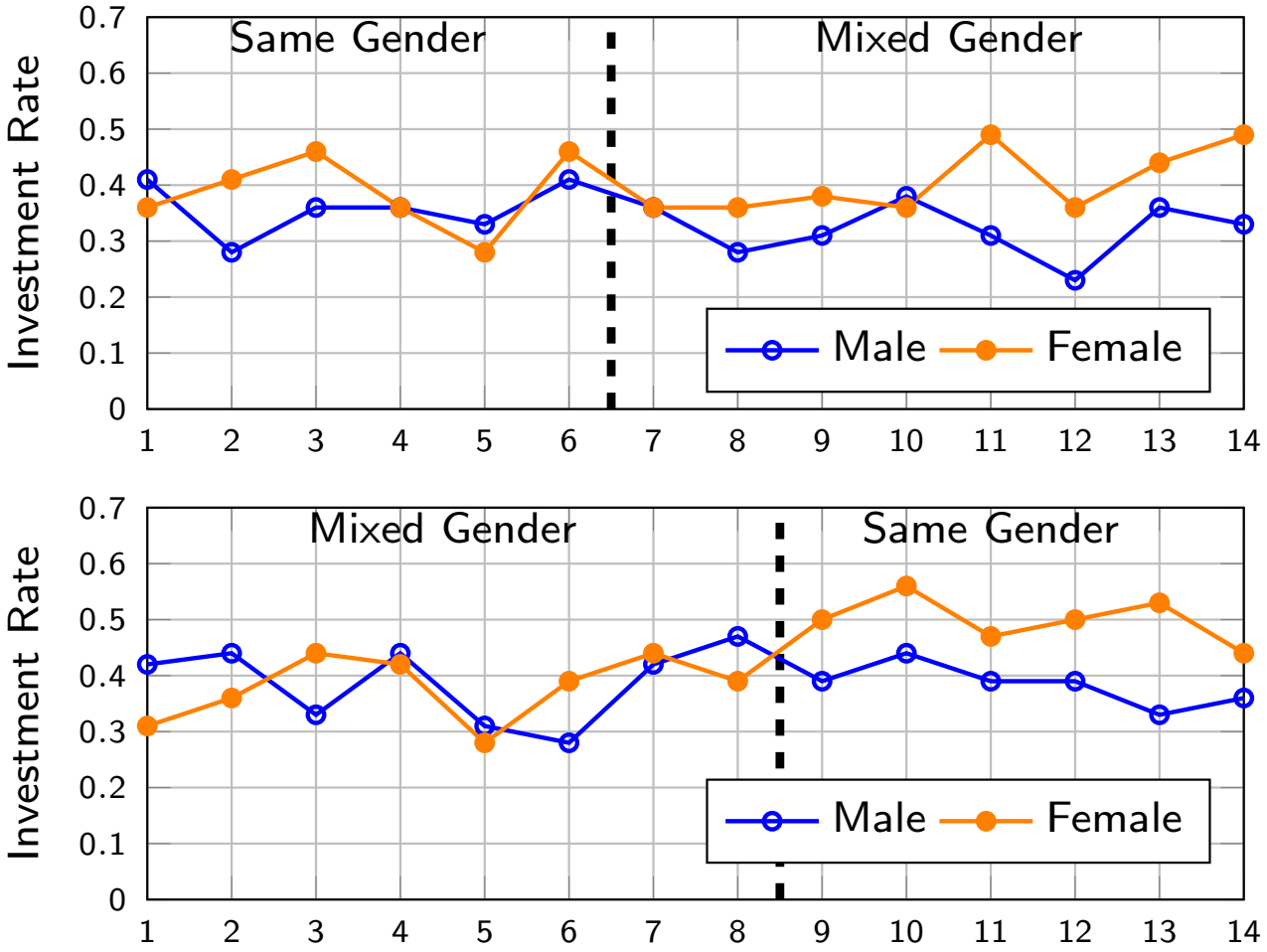


Results

- Balance tests confirm that characteristics are similar between SameFirst and MixedFirst participants.
- Groups succeed (at least one invests) in 73.81% (vs. symmetric Nash equilibrium: 54% and Babcock et al.: 84.2%)

Results

Overall, women do not volunteer for LPTs at higher rates than men in mixed-gender environments.



Investment Rate over Round by Gender, *SameFirst* and *MixedFirst* Treatments

Results

Women experiencing in same-gender settings tend to volunteer for LPTs more when placed in mixed-gender groups. (See (6))

	Pooled Data (All Rounds)		Same Environment Only		Mixed Environment Only	
	(1)	(2)	(3)	(4)	(5)	(6)
SameFirst	-0.0508	-0.0915	-0.0591	-0.0958	-0.1022	-0.1409*
	(0.525)	(0.166)	(0.599)	(0.348)	(0.235)	(0.053)
Female	0.0428	-0.0106	0.1131	0.0566	-0.0103	-0.0603
	(0.548)	(0.867)	(0.211)	(0.490)	(0.888)	(0.365)
SameFirst×Female	0.0185	0.1000	-0.0814	-0.0090	0.0956	0.1806*
	(0.859)	(0.273)	(0.478)	(0.932)	(0.383)	(0.057)
Round	0.0030	0.0032	-0.0042	-0.0037	0.0056	0.0058
	(0.277)	(0.250)	(0.595)	(0.631)	(0.244)	(0.227)
Key Controls						
selfPressure	-	0.0970***	-	0.1192***	-	0.0806***
		(0.000)		(0.000)		(0.000)
volunteer	-	0.0503**	-	0.0405	-	0.0585**
		(0.022)		(0.149)		(0.015)
sameGenExp	-	-0.0184	-	-0.0491**	-	0.0044
		(0.255)		(0.012)		(0.794)
Observations	2100	2100	900	900	1200	1200

Notes: SameFirst is a dummy indicating whether the same-gender environment games were played first. The table presents marginal effects of Probit regression. *p*-values based on clustered standard errors are reported in parentheses.

Results

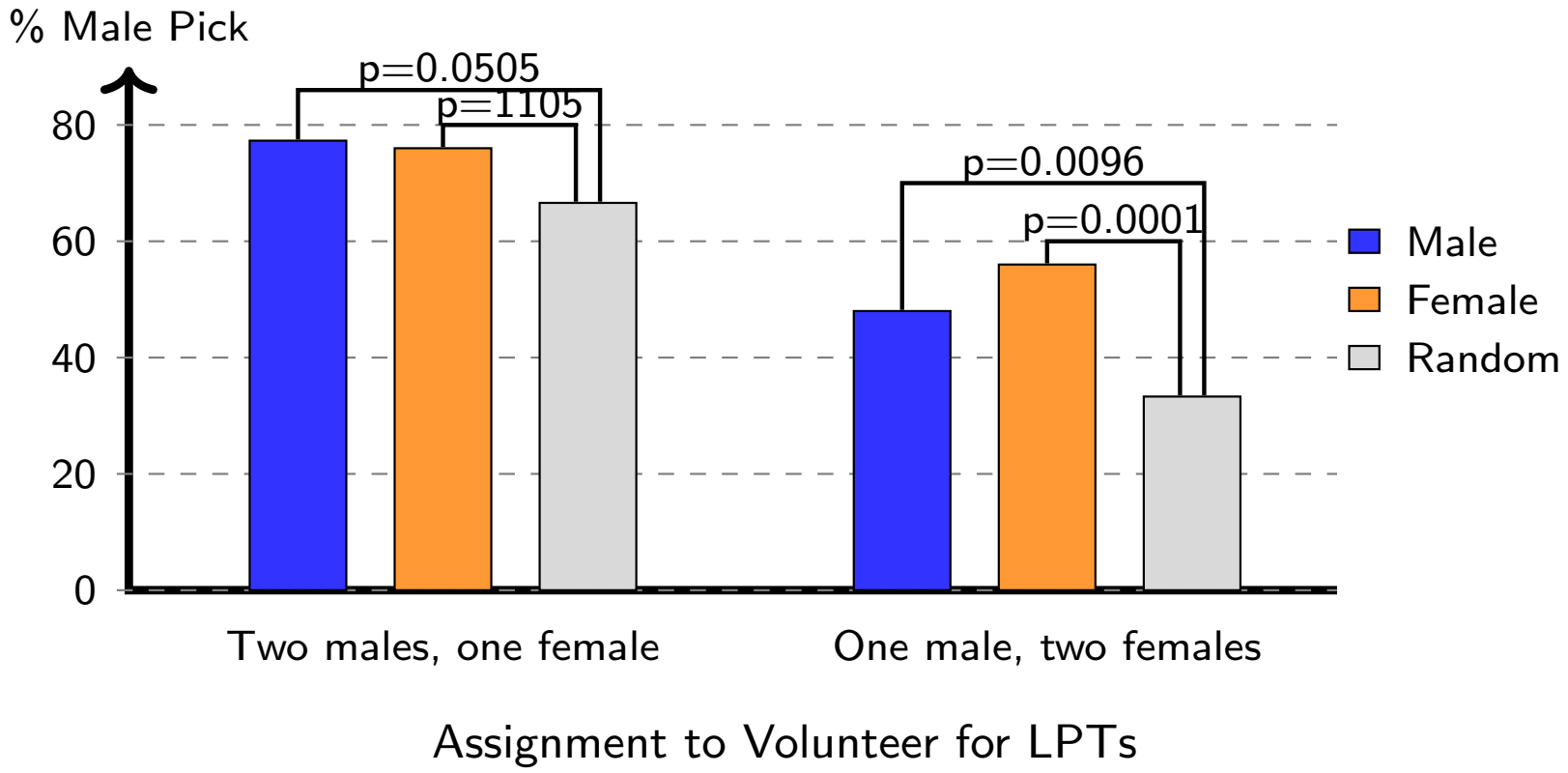
Women with single-sex schooling backgrounds are more likely to volunteer for LPTs than women from coeducational schools, particularly within mixed-gender environments.

(Female only)	Pooled Data (All Rounds)		Same Environments Only		Mixed Environments Only	
	(1)	(2)	(3)	(4)	(5)	(6)
SameFirst	-0.0049	0.0113	-0.0943	-0.1085	-0.0363	0.0088
	(0.953)	(0.884)	(0.486)	(0.439)	(0.699)	(0.925)
Girl School	0.1180	0.1371*	0.1305	0.1126	0.1093	0.1612**
	(0.145)	(0.074)	(0.268)	(0.313)	(0.170)	(0.041)
Interaction	-0.0637	-0.0401	-0.1162	-0.0445	-0.0259	-0.0412
(SameFirst × Girl School)	(0.631)	(0.747)	(0.447)	(0.767)	(0.851)	(0.749)
Round	0.0089**	0.0091**	-0.0042	-0.0041	0.0122*	0.0121*
	(0.015)	(0.016)	(0.723)	(0.726)	(0.072)	(0.074)
Key Controls						
selfPressure	-	0.0770***	-	0.0926***	-	0.0666**
		(0.005)		(0.004)		(0.019)
Observations	1,050	1,050	450	450	600	600

Notes: Dependent variable: individual investment decision (1–invest, 0–don't invest). Survey controls include selfPressure, sameGenExp, concernOthers, volunteer, leader, socialConcern, nonConfirmity. The table presents marginal effects at the means. Standard errors are clustered at the individual level. *p*-values are reported in parentheses.

Results

Subjects, as supervisors, tend to assign an LPT to a male member in a mixed-gender group.



Previous studies report that females are more requested to do LPTs (Babcock et al., 2017; Bircan et al., 2025), with expecting that females would accept the request. We find the opposite.

Summary of the Findings

We examine if experiences in same-gender settings shape willingness to volunteer for LPTs in mixed-gender groups.

- Exposure to same-gender environments reduces overall volunteering when later interacting in mixed-gender groups.
- Conditional on such exposure, women are more likely than men to volunteer in mixed-gender settings.
- This gender difference is amplified among women with single-sex schooling backgrounds.
- Importantly, these patterns arise *without explicit expectations that women should perform LPTs*.

Discussions

- Taken together, the findings indicate persistence of behavioral responses, but not persistence of gendered social norms. That is, women do LPTs more in mixed-gender settings, but this is not driven by internalized social expectations.
- This finding directly contradicts the notion that women volunteer more because they expect to be asked, sanctioned, or judged more harshly for refusing.
- We claim it suggests a different behavioral driver: the motivation to resolve strategic uncertainty. Perhaps women do LPTs more not because it is expected for them to do so, but because they want to avoid coordination failure due to greater unpredictability regarding others' actions in mixed-gender groups (Eckel and Füllbrunn, 2015; Cason et al., 2022).

Discussions: Supplementary Evidence

Marginal Effects on Group Success Rate (Coordination)

	Pooled Data (All Rounds)		Same Environment Only		Mixed Environment Only	
	(1)	(2)	(3)	(4)	(5)	(6)
SameFirst	-0.0148	-0.0307	0.0862	0.0698	-0.0605	-0.0756
	(0.690)	(0.349)	(0.263)	(0.362)	(0.242)	(0.130)
Female	0.0668**	0.0456*	0.1005**	0.0906**	0.0406	0.0099
	(0.030)	(0.099)	(0.029)	(0.041)	(0.307)	(0.792)
Interaction	-0.0618	-0.0261	-0.1615**	-0.1370*	0.0079	0.0509
(SameFirst × Female)	(0.227)	(0.582)	(0.036)	(0.064)	(0.891)	(0.336)
Round	-0.0010	-0.0009	0.0052	0.0050	0.0003	-0.0001
	(0.684)	(0.698)	(0.502)	(0.519)	(0.952)	(0.987)
Key Controls						
selfPressure	-	0.0368***	-	0.0388***	-	0.0356***
		(0.001)		(0.008)		(0.006)
Observations	2100	2100	900	900	1200	1200

Notes: Dependent variable: group success (1-at least one invests, 0-otherwise). The table presents marginal effects of Probit regression. P-values based on clustered standard errors are reported in parentheses.

(Although women do not invest at higher rates on average,) groups composed of female-only participants achieve significantly higher coordination success. This pattern is more pronounced following exposure to mixed-gender environments.

Discussions: Policy Implications

Policy suggestions would differ by the main reason for women's LPT volunteering.

- If the gender gap in LPT volunteering were driven solely by unjustifiable social expectations, appropriate interventions might focus on individual-level behavioral adjustments aiming to ultimately alter social norms.
- If women's volunteering is a rational response to strategic uncertainty in coordination-game-like settings, focusing solely on individual behavior is potentially destructive.
- Our findings support a shift away from individual behavioral adjustments and toward structural, institutional interventions (such as rotational assignment) that reduce the strategic uncertainty altogether.

Discussions: Limitations

Several limitations warrant mention.

- The laboratory setting abstracts from explicit task requests, repeated interactions, and hierarchical workplace relationships.
- Although the evidence is consistent with a strategic-uncertainty mechanism, uncertainty itself is not directly manipulated in our setting. Our claims are suggestive.